

Equipment Borrowing Monitoring System

ASP.NET Core MVC Laboratory Exercise Specification

Objective

Develop an ASP.NET Core MVC application to monitor equipment borrowing and returns using Model Binding, Cookie Authentication, Data Annotation Validation, Repository Pattern (in-memory), Razor Views, Bootstrap, and CRUD.

Business Scenario

An IT department lends equipment to students and faculty. Staff register borrowing transactions, monitor active loans, and record returns.

Learning Objectives

MVC, Model Binding, DTOs, Validation, Cookie Authentication, Authorization, Repository Pattern, CRUD, Search, Bootstrap.

Functional Requirements

Register User, Login, Logout, Register Borrowing, View Monitoring List, Edit Borrowing, View Details, Return Equipment, Search.

Borrowing Fields

Transaction Number, Borrower Name, Borrower Type, Student/Employee ID, Department/Course, Equipment Name, Equipment Category, Quantity, Borrow Date & Time, Expected Return Date, Purpose, Notes.

Monitoring

Display Transaction Number, Borrower, Equipment, Quantity, Borrow Date, Return Date, Status, Actions.

Status

Borrowed, Returned.

Validation

Use Required, Range, StringLength, Display, DataType and validate ModelState.

Authentication

Public: Login/Register. Protected: Equipment pages using Cookie Authentication and [Authorize].

Entities

User: Id, FirstName, LastName, Email, Username, Password. EquipmentBorrowing: Id, TransactionNumber, BorrowerName, BorrowerType, BorrowerId, DepartmentOrCourse, EquipmentName, EquipmentCategory, Quantity, BorrowDateTime, ExpectedReturnDate, ActualReturnDateTime, Status, Purpose, Notes.

Repositories

UserRepository and EquipmentBorrowingRepository using static List.

Controllers & Views

Controllers: HomeController, AccountController, EquipmentBorrowingController. Views: Login, Register, Index, Create, Edit, Details, Return.

Submission

Complete MVC solution with authentication, CRUD, validation, search, Bootstrap UI, repository pattern, and clean code.